CPE 301.1001 Embedded Systems Design

Final Project Part 3, by Kevin Takamura

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1 Project Introduction

Public transport in the United States—especially Nevada—is overshadowed by the vast and efficient networks in other countries like Japan and China. Through my project, I aim to make public transit in Nevada more accessible through an RFID card system, akin to the Bay Area Clipper Card, named the AceCard. Riders would be able to transfer their existing paper or digital passes from any Nevada transportation agency onto a durable physical RFID card that they can tap on a card reader when boarding supported busses or other transportation methods, simplifying the boarding process and unifying Nevada's various transit systems. This would potentially allow passengers to board from the rear doors—particularly on larger Vegas buses—also decreasing the amount of time each stop takes, increasing overall system efficiency. An embedded system would be best in this scenario since its application requires a compact, low-power, specialized low-computing system that does not use much energy when installed on buses.

2 Proposed System

Please see the next page.

2.1 System Schematic

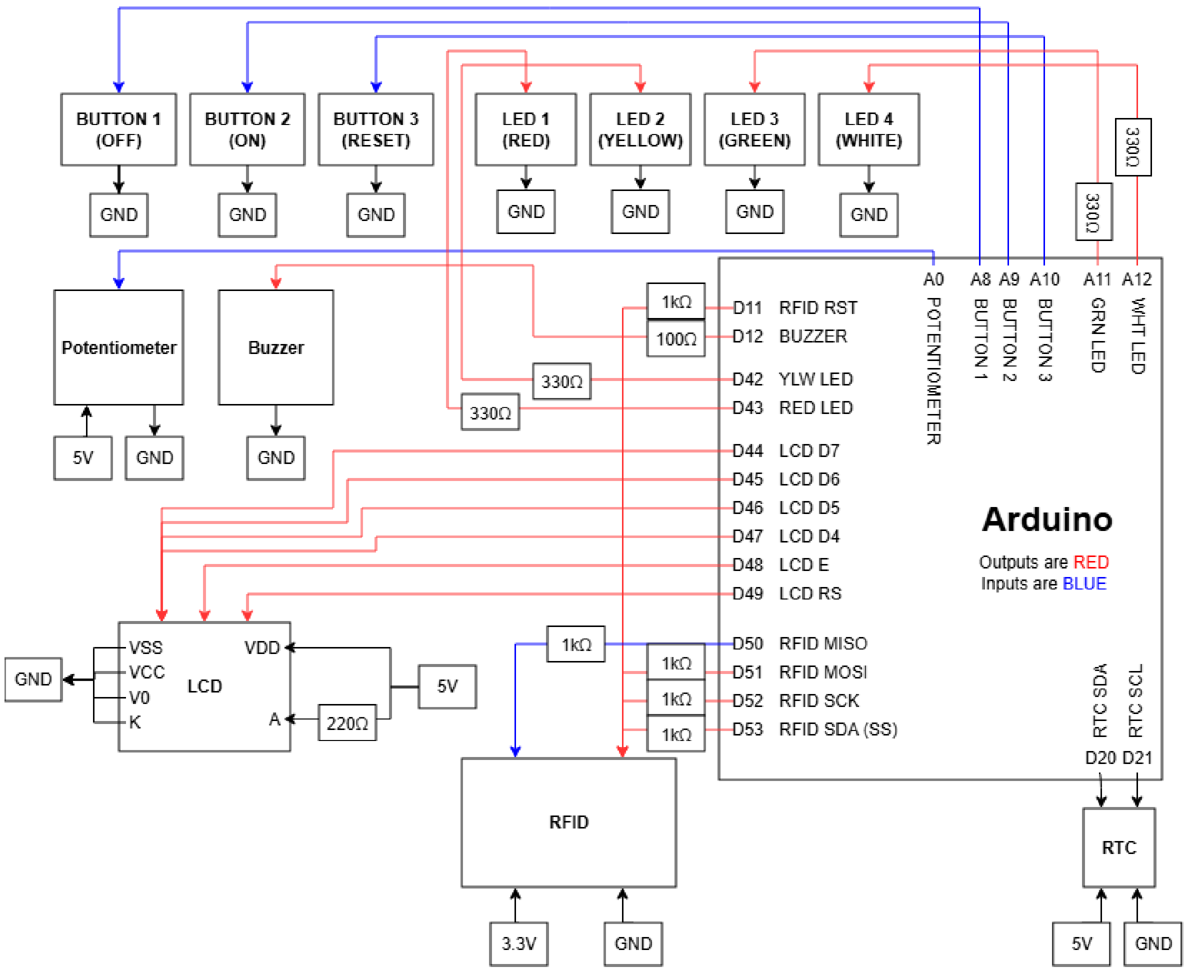

Figure 1: A full schematic diagram of the system. A high-resolution version can be found here: https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=schematic.drawio&dark=auto#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1jN50G8QXKhaJM2uqo1-Kg3qDOczfs8aL%26export%3Ddownload

2.2 Component List

- One Arduino Mega 2560 + jumper wires and a breadboard
- One 100Ω resistor for the buzzer
- One 220Ω resistor for the LCD backlight
- Four 330Ω resistors for each LED
- Five 1kΩ resistors for each RFID data line
- Three buttons
- Four LEDs (One each in red, yellow, green, and white)
- One DS1307 Real Time Clock (RTC) module
- One 16x2 LCD display

- One RC522 RFID transceiver
- One potentiometer
- One passive buzzer

3 System Behavior

3.1 Module Design

The AceCard system is divided into three modules. The first module encapsulates the buttons and potentiometer; in other words, the simple inputs. The second module contains the LEDs and buzzer; in other words, the simple outputs. The third module is dedicated to the LCD, RFID transceiver, and RTC; in other words, the complex devices. The buttons control the state of the device and allow it to switch between OFF, IDLE, and out of ERROR. When not in IDLE, the RFID transceiver and potentiometer stop polling. The potentiometer controls the number of credits deducted from the AceCard presented to the RFID transceiver when tapped. The LEDs solely represent the state of the device; white is OFF, green is IDLE, yellow is ACTIVE, and red is ERROR. The buzzer chirps a different tune based on if the card did or did not have enough credits to complete the transaction and buzzes constantly when in ERROR. The LCD displays the current state along with helpful messages or the current fare when in IDLE. The RFID transceiver communicates only with the Arduino but can change the state of the system from IDLE to ACTIVE when a card is presented. The RTC is used only for time logging in the serial monitor and does not interact with any components other than the Arduino.

3.2 State Diagram

Please see the next page.

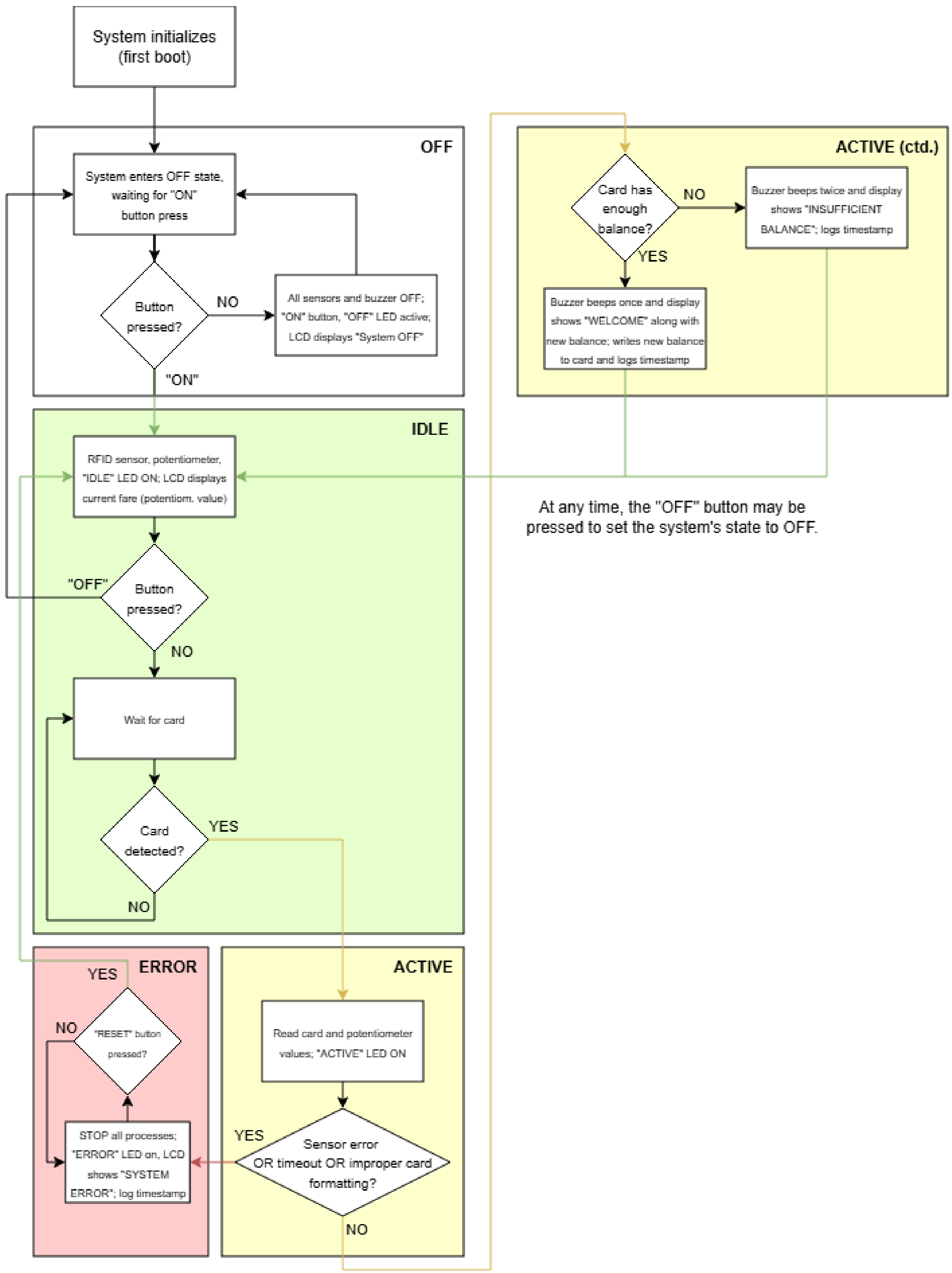

Figure 2: A flowchart illustrating the processes the system follows under both normal and abnormal conditions. Please note that at first boot, the system checks the RFID and RTC modules. If either fails to communicate, the system immediately enters ERROR. A high-resolution version can be found here: https://app.diagrams.net/?title=SystemFlowchart.drawio&dark=auto#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1t9QgTQ27HkL903imxoacWs0nhcaFiw8o%26export%3Ddownload

4 Result

A video link is attached in the Github repository. Due to LaTeX formatting issues, the serial monitor screenshot is at the start of page seven, captioned ”Figure 7”.

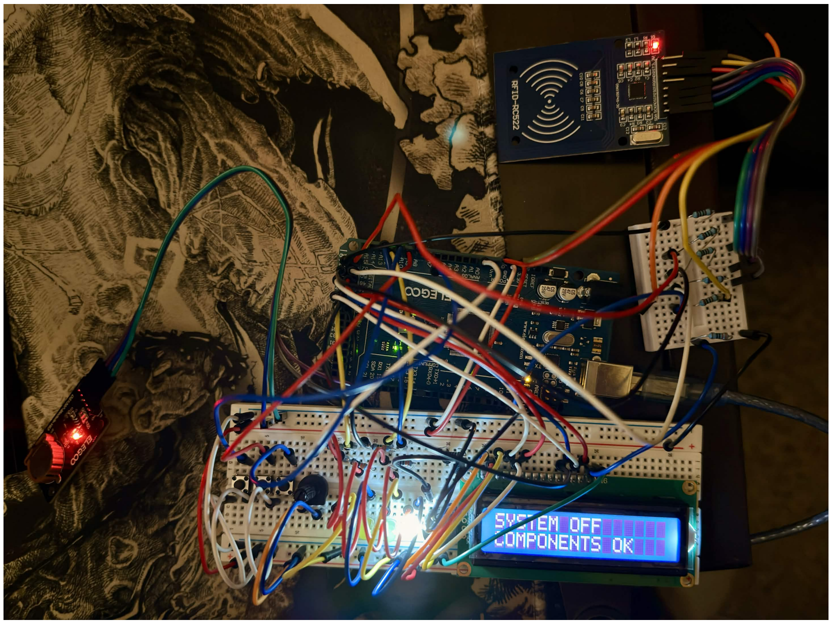

Figure 3: A full circuit image with the system in STATE 0: OFF. The RTC module is the red LED on the left.

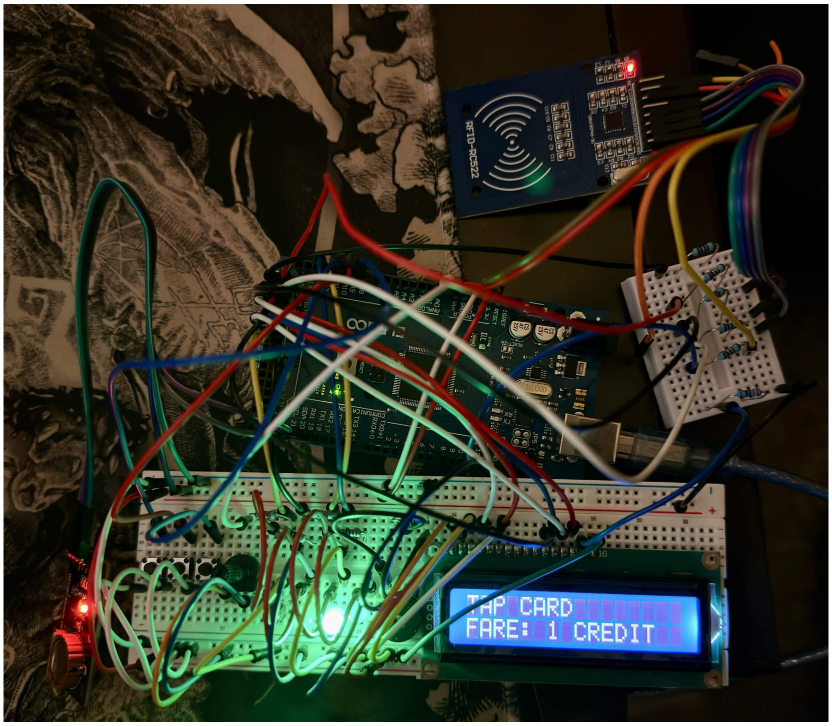

Figure 4: A photo of the system in STATE 1: IDLE.

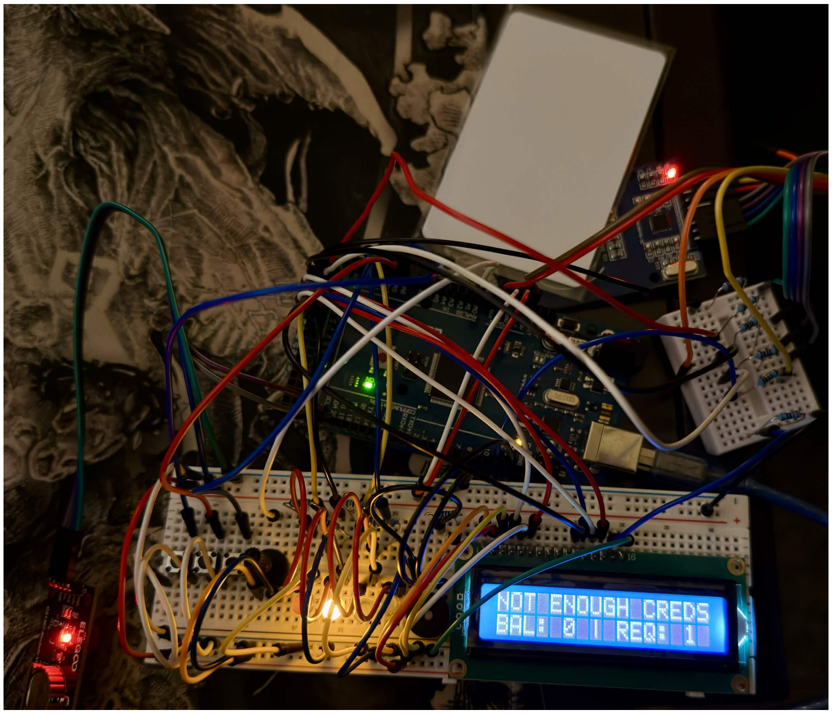

Figure 5: A photo of the system in STATE 2: ACTIVE with a genuine AceCard on the RFID transceiver.

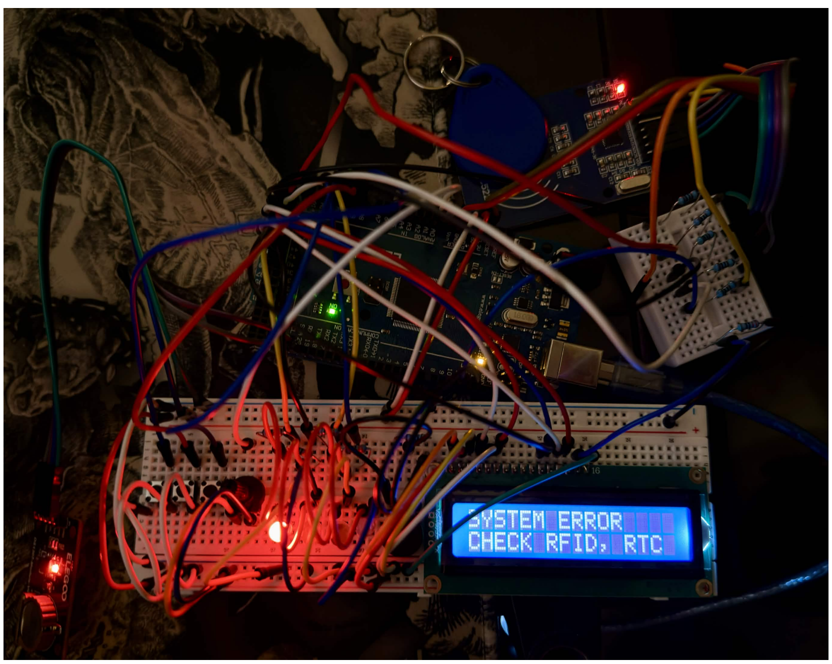

Figure 6: A photo of the system in STATE 3: ERROR with a counterfeit AceCard on the RFID transceiver.

5 Environmental & Energy Impact

Please disregard figure 7; it is here as a result of LaTeX formatting issues. The section text starts at ”5.1 Electrical and Design Safety”.

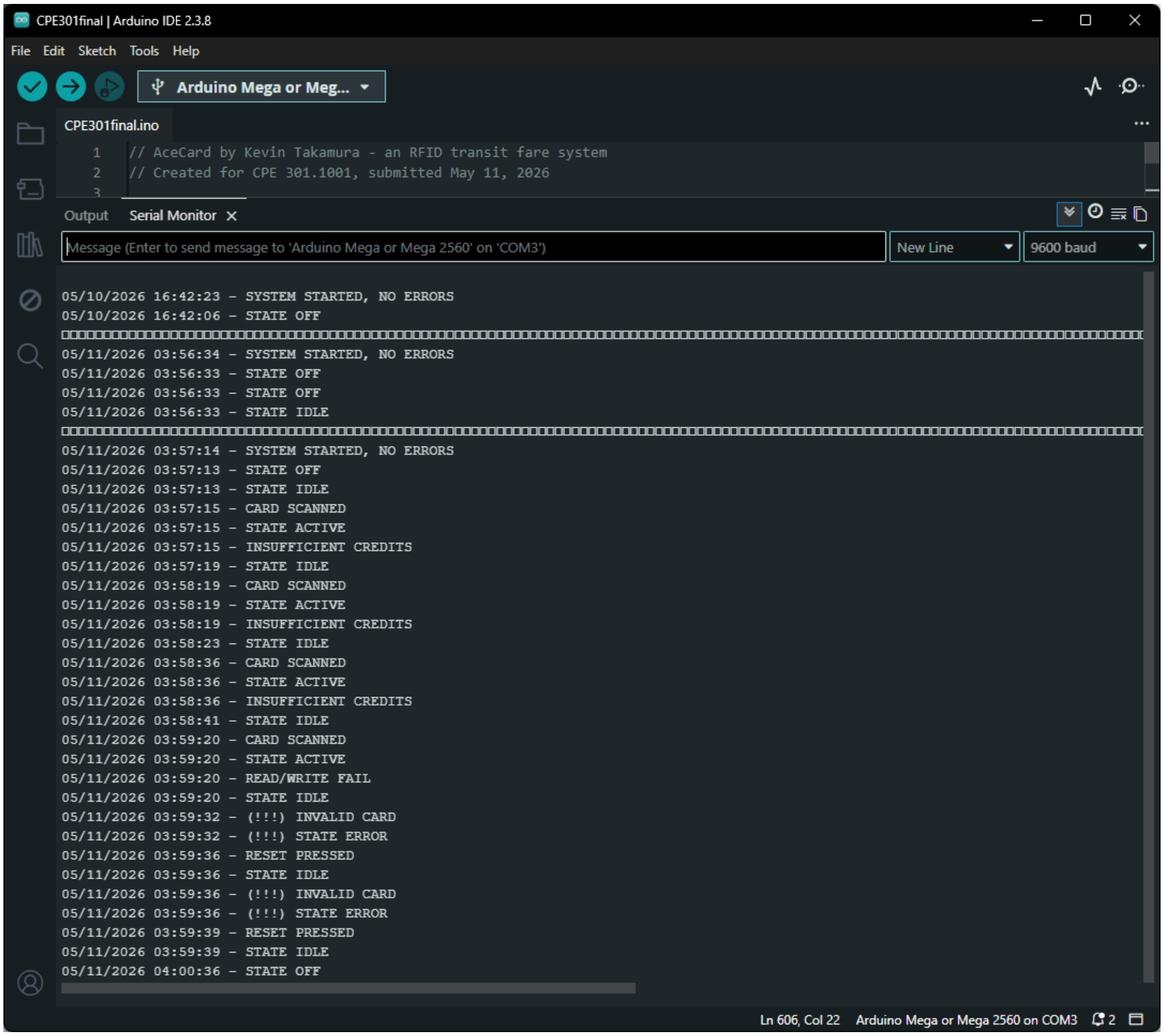

Figure 7: A screenshot of the serial monitor output of the system. The "INVALID CARD" errors were a response to the counterfeit AceCard.

5.1 Electrical and Design Safety

Considering the small size of the transceiver, the system should use minimal energy while powered on. Although it would be powered on for long periods of time, bus operators (or a door interlock) will only turn on the transceiver under certain conditions (like the bus being stopped with the respective door open), thereby reducing the amount of power it uses. Since similar fare systems are present on some bus platforms in other regions, implementing the system while keeping electrical safety in mind should not be a significant issue since bus electrical systems are fairly simple and robust. Electrical protections will be in place to ensure the safety of the system's electronics, and even if the system fails, it would enter its ERROR state, flashing a constant red light, beeping with a constant tone, and displaying an error code in the serial monitor without posing any risk (since the bus can continue operating normally; albeit, riders would need to use alternative methods of providing a fare, or rides would temporarily be free). Additionally, the system can check if the RFID chip it detects is formatted in a way consistent with AceCard, and when a credit is being deposited or withdrawn to/from a balance, the system waits for the card to be removed before deducting from the balance again, so there is no risk of error.

5.2 Sustainability

AceCards are meant to be reusable and long-lasting, eliminating the need for paper tickets and printers. This provides opportunities for riders to buy AceCards in materials like steel or titanium instead of plastic or printed with custom designs to raise money for local transit agencies (so they can make transit even better!) and other organizations. This would reduce the waste that paper tickets create while being versatile enough to be used for different applications in the future. For example, if transit were to be free, the cards could be used to track how many times a rider has ridden public transport so they can get rewards (similarly modeled after RTC Southern Nevada's ClubRide program). Additionally, riders with compatible devices could add the card to their mobile wallets, just like Clipper Card!

5.3 Affordability

All components used in this project are included in the course-provided kit, meaning that expanding it to a real-world application would be inexpensive and easy. Despite a higher initial cost for riders (something like $5 for a new card), the card should last years before needing a replacement, at which point it can be recycled. The housings to make the components look ergonomic and tasteful could draw inspiration from Clipper and use 3D-printed parts, further reducing the complexity and price of the completed build.

5.4 Accessibility

AceCard could be purchased at any major transit hub in Nevada and reloaded at select bus stops and on board buses. All riders, including those with disabilities, should have no trouble using AceCard since using it on board buses is as simple as holding the card against the reader and waiting for a buzzer. There will be no other way for riders to interact with the device; only drivers, customer service agents, or kiosks are allowed to use the ON, OFF, or RESET buttons or fare-controlling potentiometer.

6 Conclusion

In summary, the AceCard system is fully functional. It can successfully detect if the RFID or RTC modules are non-functional when the system first boots (in other words, when it receives power) and can deter malicious actors from using fake AceCards since the system checks the card's ID at every scan and alerts the driver when a "bad" card is scanned. The potentiometer accurately modifies the fare and the display refreshes once a minute with its value, and the buttons and LEDs successfully function as state switchers/lights. Of course, the RFID chip works properly and communicates with RFID cards without errors, deducting the selected fare with ease. I was unable to add more complexity due to time constraints. Among the unrealized additions, I wanted to add a system to see if the RFID module was still responding once a minute, add a button so that fare is added and not subtracted from the card, and replace the buzzer library functions with timers. Some system limitations include the physical footprint of the system and RFID data encryption. The system takes up an entire breadboard and then some, so size may be a problem unless soldering is used. Furthermore, in its current state, the RFID system only works up to 255 credits stored on a specific bit and can only continue to fare calculations when a specific RFID ID is detected. Additionally, the data is not encrypted, so anyone could simply write to the card using their own RFID writer. In a real-world application of the system, the RFID data would be encrypted and the system would be placed next to the driver so they can interact with the buttons and potentiometer. There would also be a way to read from the serial monitor so the driver/operator can read any necessary logs. Generally, however, I am extremely satisfied with my finished product.